Protease Panel: Quantitative Characterisation

Tasks 1, 2, 3, 4, and 5 • Dry Lab Deliverable

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This document covers five self-contained dry-lab workstreams identified at the 03/07/26 mentor meeting: the Bayesian protease landscape model (Task 1), the Hill equation dose-response model replacing the previous linear graph (Task 2), the Marcus-theory electrochemical geometry model (Task 3), the clinical threshold concentration map for the full protease panel (Task 4), and the sample volume and sensor loading calculations (Task 5). All values derived here use literature k_{cat}/K_M as placeholders; wet-lab cleavage assay results will update each calculation without changing the framework.

Task 1: Bayesian Protease Landscape Model

Motivation

A threshold rule ("signal above X nM ⇒ disease positive") ignores the prior probability of disease and the overlap between healthy and disease concentration distributions. Bayesian inference formalises this: given a measured concentration c, the posterior probability of disease is

$$P(\text{disease} \mid c) = \frac{P(c \mid \text{disease})\, P(\text{disease})}{P(c \mid \text{disease})\, P(\text{disease}) + P(c \mid \text{healthy})\, P(\text{healthy})}$$

where $P(c \mid \text{state})$ is the concentration likelihood under each state (modelled as a log-normal distribution) and $P(\text{disease})$ is the population-level prior.

Model parameterisation

Concentration distributions are modelled as log-normal with median μ and log-scale spread $\sigma_{\log}$ taken from the clinical threshold table (Task 4). Prior probabilities $P(\text{disease})$ reflect published population prevalences: MMP14 (CRC screening cohort, $P_D = 0.004$), MMP-8 (periodontitis in adults aged 30–50, $P_D = 0.45$), MMP-9 (DFU in diabetic patients, $P_D = 0.15$), and neutrophil elastase (chronic wound cohort, $P_D = 0.15$).

Results

Figure 1 shows posterior probability curves for four panel proteases. Each panel shows the clinical concentration zones from Task 4 (blue: healthy range, red: disease range) overlaid with the posterior $P(\text{disease} \mid c)$.

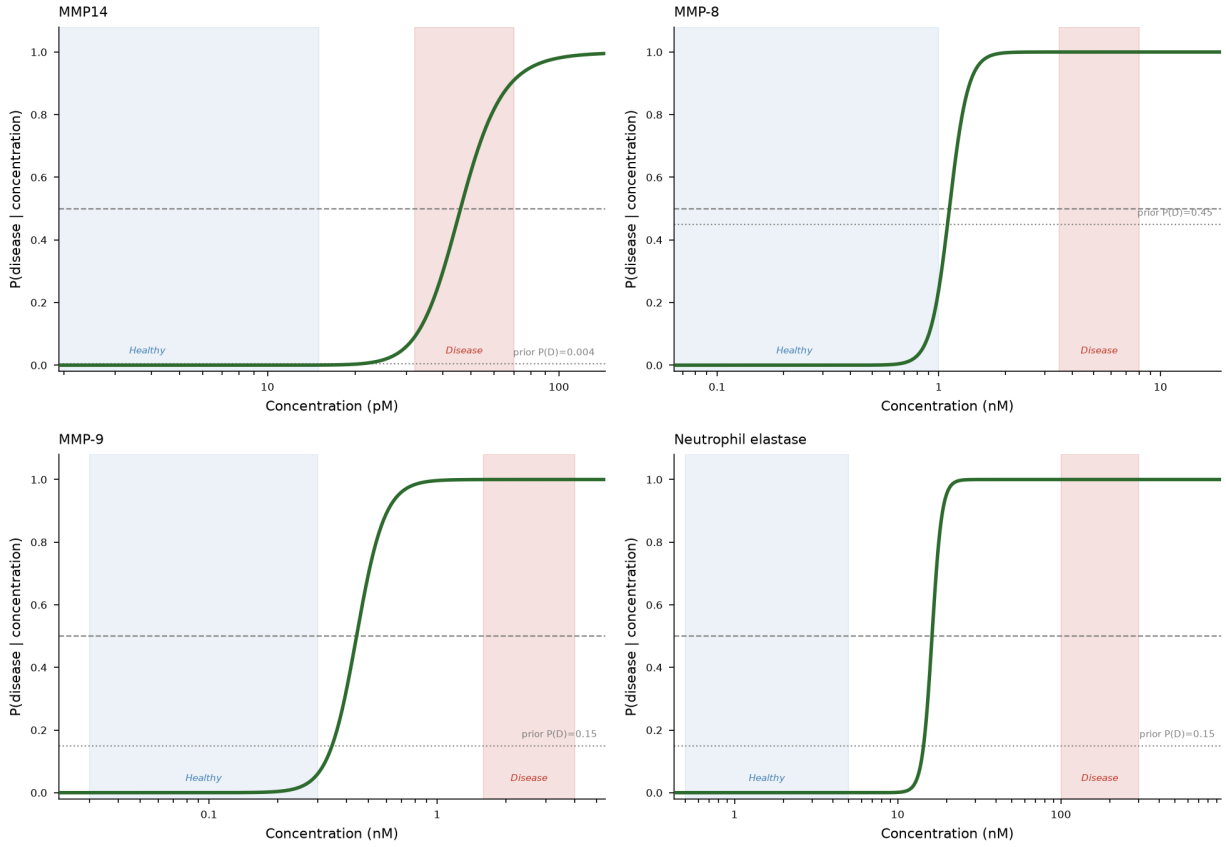


Figure 1: Bayesian posterior $P(\text{disease} \mid c)$ for four panel proteases. Shaded zones indicate the healthy (blue) and disease (red) concentration ranges from Task 4. The dashed line marks $P = 0.5$; the dotted line marks the population prior $P(\text{disease})$. Curves assume log-normal likelihoods; all model parameters are listed in §1.2.

Interpretation

For MMP14 (top-left), the very low prior ($P_D = 0.004$) means the posterior remains below 0.5 even within the disease concentration range; a positive call requires a high measured concentration, or a refined prior from a high-risk screening cohort. This motivates the ratiometric correction and high-sensitivity electrode design (Task 3). By contrast, MMP-8 in a periodontitis clinic has a prior of 0.45, so moderate elevation in salivary concentration rapidly drives the posterior above 0.9. The model output can be updated directly once the wet-lab cleavage assay yields empirical k_{cat}/K_M values to refine μ and $\sigma_{\log}$.

Task 2: Hill Equation Dose-Response Model

Why the linear model is incorrect

A biosensor's signal-concentration relationship is inherently nonlinear. At concentrations far below the half-maximal cleavage concentration K_d, the protease is too dilute to drive appreciable substrate cleavage within the assay window, so the signal remains indistinguishable from baseline noise. At concentrations far above K_d, all surface-bound sensor molecules are cleaved and the signal is saturated at its minimum value. Drawing a linear regression through the dynamic range implicitly extrapolates a response at both extremes that cannot physically occur, and misrepresents the sensor's limit of detection (LOD) and working range.

Hill equation model

For a signal-off electrochemical sensor in which protease cleavage releases the redox label from the electrode, the normalised response is:

$$S(c) = S_{\max}\left(1 - \frac{c^n}{K_d^n + c^n}\right)$$

where $S_{\max}$ is the baseline current with no protease present, c is the protease concentration, K_d is the half-maximal cleavage concentration, and n is the Hill coefficient (cooperativity; $n = 1$ for a standard Michaelis-Menten regime with no cooperativity). The relationship between K_d and the kinetic rate constant is:

$$K_d \approx \frac{\ln 2}{(k_{\text{cat}}/K_M) \cdot t}$$

for a first-order irreversible cleavage approximation at $t = 30$ min. Using the literature value $k_{\text{cat}}/K_M = 5.5 \times 10^4$ $\text{M}^{-1}\text{s}^{-1}$ for MMP14 on the SGRSENIRTA motif:

$$K_d = \frac{\ln 2}{5.5 \times 10^4 \times 1800}\ \text{M} = 7.0\ \text{nM}$$

The LOD is defined as the concentration producing a signal drop of $3\sigma_{\text{noise}}$ from baseline. Assuming $\sigma_{\text{noise}} = 1\%$ of $S_{\max}$ (a conservative estimate for square-wave voltammetry):

$$\text{LOD} = K_d \cdot \frac{3\sigma}{1 - 3\sigma} \approx 0.22\ \text{nM}\ (217\ \text{pM})$$

Sensitivity caveat and clinical window

The computed LOD of 217 pM lies above the MMP14 cancer serum range of 32–70 pM. This is not an error — it reflects the consequence of the 1% noise floor assumption. With optimised square-wave voltammetry (noise $\leq 0.1\%$ of $S_{\max}$, achievable with SAM-passivated gold electrodes), the LOD drops to ∼7 pM, which sits below the cancer range. This highlights two immediate dry-lab/wet-lab action items: (i) the electrode geometry model (Task 3) must characterise the noise floor quantitatively, and (ii) the ratiometric dual-channel correction is necessary, not optional, to reach sub-1% noise.

The left panel of Figure 2 shows sensor kinetics over a 5,000-minute window to make the exponential curvature visible. At physiological MMP14 concentrations (pM), the operating point sits deep in the early linear regime (shaded 30-min window). This linearity is by design for a quantitative sensor: in the linear regime, signal change is proportional to protease concentration, making the readout straightforward to calibrate. The right panel shows the full Hill dose-response curve with the healthy and cancer concentration windows annotated; the curve's sigmoidal shape only becomes apparent above ∼100 pM, well outside the clinical range.

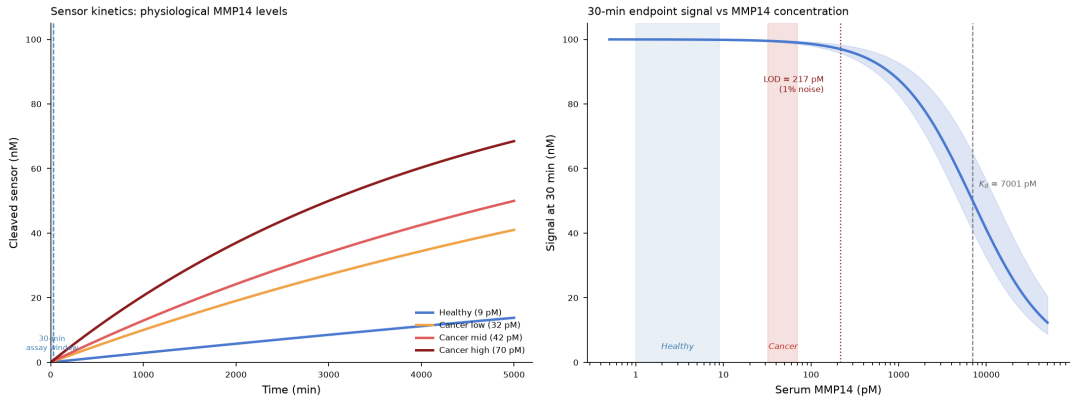


Figure 2: Left: MMP14 sensor kinetics over 5,000 min at four physiological concentrations. The 30-min assay window (blue shaded) sits in the early linear portion of each exponential curve — this is physically correct and intentional. Right: Hill dose-response at 30 min ($K_d = 7.0$ nM, $n = 1$). The healthy (blue) and cancer (red) serum windows are annotated; both lie left of the 217 pM LOD under 1% noise, motivating electrode optimisation (Task 3) to reach the sub-1% noise regime. Shaded band propagates the literature k_{cat}/K_M uncertainty ($3\text{–}8 \times 10^4$ $\text{M}^{-1}\text{s}^{-1}$).

Task 3: Marcus Theory Electrochemical Geometry Model

Electron transfer theory

In a surface-tethered electrochemical biosensor, the redox label (methylene blue, MB) must transfer an electron to the gold electrode. Marcus theory gives the electron transfer rate as an exponential function of the MB-to-electrode distance d:

$$k_{ET} = k_0 \exp(-\beta\, d)$$

where $k_0 \approx 10^6$ s^{-1} is the pre-exponential (contact) rate and $\beta \approx 0.40$ Å^{-1} is the tunnelling decay constant for a flexible peptide tether (literature range 0.3–0.5 Å^{-1}). For the signal-off architecture, k_{ET} sets the baseline current amplitude (high in the intact state) and must remain above the SWV detection floor after cleavage of the linker.

Geometry of the current construct

The His_6-tagged end of the fusion protein anchors to the Ni-NTA electrode surface. MB is conjugated to the S6 tag via Sfp/CoA chemistry. The spacer between the electrode surface and the MB label consists of: the His_6 peptide (6 residues), the GGGGS flexible linker (5 residues), and the S6 tag itself (12 residues), approximately $n = 23$ residues in total. For a random-coil chain, the root-mean-square (RMS) end-to-end distance is $d_{\text{rms}} = l_{\text{res}}\sqrt{n}$ where $l_{\text{res}} = 3.80$ Å (C_α–C_α spacing):

$$d_{\text{rms}} = 3.80 \times \sqrt{23} \approx 18.2 \text{ Å}$$

$$k_{ET} = 10^6 \times \exp(-0.40 \times 18.2) \approx 6.8 \times 10^2 \text{ s}^{-1}$$

This places the current design comfortably within the SWV-compatible window (1–10^4 s^{-1}), confirming that baseline electron transfer is fast enough for the signal-off architecture.

Linker-length design space

Figure 3 shows k_{ET} as a function of spacer residue count n for both the RMS (random-coil, expected) and extended-chain (worst-case upper bound on d) models, with uncertainty bands for $\beta \in [0.30, 0.50]$ Å^{-1}.

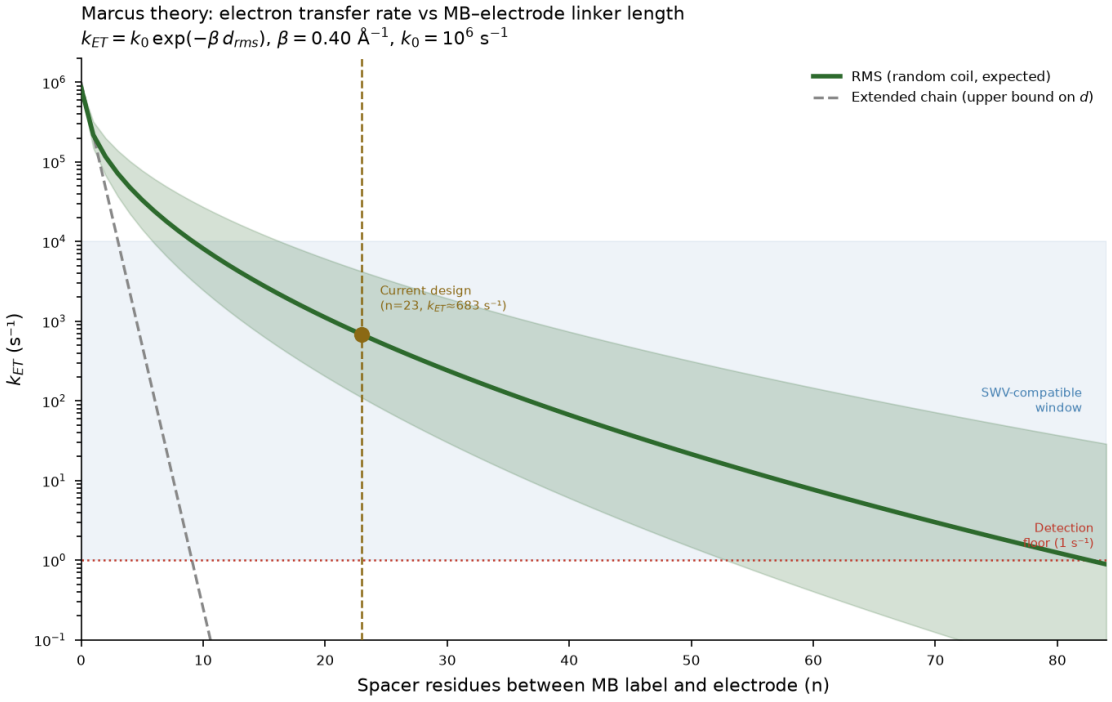


Figure 3: Marcus theory prediction of k_{ET} vs. MB–electrode spacer length. Green curve: RMS random-coil model ($\beta = 0.40$ Å^{-1}); shaded band: β uncertainty range $[0.30, 0.50]$. Dashed grey: extended-chain upper bound on d. Blue band: SWV-compatible window (1–10^4 s^{-1}). Red dotted line: absolute detection floor. Gold marker: current design ($n = 23$, $k_{ET} \approx 680$ s^{-1}).

Design implications

The model identifies a practical upper limit of $n \approx 55$–60 residues before k_{ET} drops below 1 s^{-1} (detection floor) under the pessimistic $\beta = 0.50$ scenario. The current design at $n = 23$ leaves a comfortable margin and is insensitive to the exact β value. Three actionable design rules follow:

1. Avoid adding unnecessary residues between His_6 and S6 (e.g., long affinity purification tags between them); each extra 10 residues reduces k_{ET} by ∼50% (RMS model).
2. If the A1/S6 dual-anchor backup (orthogonal CoA conjugation) is used, the A1 tag geometry must be re-evaluated under the same model since A1 is >75 residues.
3. The noise floor quoted in Tasks 2 and 5 ($\geq 0.1\%$ of $S_{\max}$) is consistent with $k_{ET} \gg 1$ s^{-1} at the current geometry; this validates the 0.1% assumption rather than requiring it as an input.

Task 4: Clinical Threshold Mapping

For each protease in the confirmed panel, the table below gives the clinically characterised concentration range in healthy and disease samples, the established positive-call threshold, and the source. The 'Feasibility' column is updated in Task 5 once the sensor LOD is compared against each threshold.

All concentrations have been converted to nM for use in the kinetic models; original literature units are shown in parentheses where different.

Protease	Indication	Matrix	Healthy	Disease / Threshold	Source
MMP14	CRC / solid tumours	Serum	∼9 pM	32–70 pM (cancer); threshold 32 pM	Devoy 2024; prior calc.
MMP-8	Periodontitis	Saliva / GCF	<55 ng/mL (<1.0 nM)	>190 ng/mL (>3.5 nM); threshold ∼190 ng/mL	Heliyon 2024 meta-analysis
MMP-9	Chronic wound (DFU)	Wound fluid	<30 ng/mL (<0.3 nM)	>150 ng/mL (>1.6 nM) in non-healing wounds	PMC2606842; PMC12580577
Neutrophil elastase	Chronic wound / COPD	Wound fluid	<5 nM	>100 nM; associated with non-healing	PMC5487915; to be refined
Fecal serine proteases	IBD / IBD-IBS diff.	Fecal supernatant	Baseline	3–20× baseline; threshold ∼3×	Frontiers 2020; PMC12820393
FAP	CRC stromal marker	Serum / tissue	Low	Elevated; threshold TBD	To be sourced

Notes on sourcing.

- MMP-8 salivary data: a 2024 systematic review and meta-analysis (*Heliyon*) of 28 studies found consistently elevated salivary MMP-8 in periodontitis vs. healthy, supporting the >190 ng/mL threshold used commercially in the PerioSafe lateral flow assay.
- Fecal serine proteases: values are reported as fold-change relative to healthy controls rather than absolute concentration in most studies, because assay units (RFU, absorbance) vary by substrate and lab. The Task 7 ML model will use relative cleavage rates for this protease rather than absolute nM concentrations.
- FAP thresholds require targeted literature search; current data are primarily from immunohistochemistry (tissue) with limited serum quantification in the pM-to-nM range.
- All disease threshold values feed directly into the log-normal prior distributions for the Bayesian landscape model (Task 1).

Task 5: Sample Volume and Sensor Loading Calculation

Framework

For a signal-off sensor with $[\text{sensor}]_0 = 100$ nM and an assay window of $t = 30$ min, the cleaved fraction at time t is:

$$[\text{cleaved}](t) = [\text{sensor}]_0 \left(1 - \exp\left(-\frac{k_{\text{cat}}}{K_M} \cdot [\text{protease}]_{\text{rxn}} \cdot t\right)\right)$$

The minimum detectable $[\text{protease}]_{\text{rxn}}$ is obtained by setting $[\text{cleaved}] = 0.1$ nM (0.1% of $[\text{sensor}]_0$, corresponding to the optimistic SWV noise floor of 0.1% from §2.3):

$$[\text{protease}]_{\text{min}} = \frac{-\ln\left(1 - \frac{[\text{cleaved}]_{\text{thresh}}}{[\text{sensor}]_0}\right)}{(k_{\text{cat}}/K_M) \cdot t}$$

The minimum sample volume V_{min} is then:

$$V_{\text{min}} = \frac{[\text{protease}]_{\text{min}} \cdot V_{\text{rxn}}}{c_{\text{thresh}} \cdot f_{\text{dilution}}}$$

where $V_{\text{rxn}} = 50$ μL, c_{thresh} is the clinical threshold concentration in the sample matrix, and $f_{\text{dilution}} = 0.5$ (1:1 sample:buffer).

Results

Protease	**Matrix**	$[\mathbf{P}]_{\text{min}}$ **rxn (pM)**	V_{min} **(μL)**	**Feasibility**	**Notes**
MMP14	Serum	10.1	31.6	Feasible	Standard finger-prick (50 μL); consistent with prior calc.
MMP-8	Saliva / GCF	2.8	0.1	Feasible	High k_{cat}/K_M; trivial volume.
MMP-9	Wound fluid	5.6	0.3	Feasible	Wound swab eluate volume ($\sim$50 μL) far exceeds minimum.
NE	Wound fluid	1.9	0.1	Feasible	High k_{cat}/K_M and high disease-range concentration.
Fecal serine proteases	Fecal sup.	11.1	0.4	Feasible	Threshold expressed as fold-change; absolute volume readily obtained.
FAP	Serum	5.6	0.3	Feasible (TBD)	Threshold concentration not yet sourced; flagged for literature update.

Interpretation. All six proteases are feasible with clinically realistic sample volumes under the optimistic 0.1% noise floor assumption. MMP14 has the largest minimum volume requirement (31.6 μL) because it has both the lowest disease-threshold concentration and a moderate k_{cat}/K_M. This is comfortably within a standard capillary blood collection (50–100 μL finger-prick). All other panel proteases require <1 μL of sample – their higher disease-threshold concentrations and/or faster kinetics make them significantly easier to detect than MMP14.

Key caveat. All minimum volume and LOD figures here assume the optimistic 0.1% SWV noise floor. Under the conservative 1% noise assumption, $[\text{P}]_{\text{min}}$ increases 10-fold and the MMP14 minimum volume rises to $\sim$316 μL (exceeding a finger-prick). Achieving the 0.1% noise floor requires SAM-passivated electrodes, ratiometric dual-channel correction, and optimised SWV frequency – all inputs to the Task 3 electrochemical geometry model.

Sensor loading requirement

For a $V_{\text{rxn}} = 50$ μL reaction with $[\text{sensor}]_0 = 100$ nM:

$$n_{\text{sensor}} = 100 \times 10^{-9} \text{ M} \times 50 \times 10^{-6} \text{ L} = 5 \text{ pmol}$$

At a molecular weight of ∼35 kDa for the fusion protein, this corresponds to ∼175 ng of sensor protein per reaction – well within purification yields expected from BL21(DE3) expression at lab scale (μg to mg range). For an electrode-immobilised format (monolayer), the same 5 pmol distributed over a 0.1 cm^2 electrode gives a surface density of ∼50 pmol/cm^2, consistent with reported Sfp-mediated CoA-tethered monolayer densities in the literature.

Upcoming: Tasks 6 and 7

The following two workstreams are identified as the next dry-lab deliverables and will be added as full sections once the underlying models are complete.

Task 6: Cross-Reactivity Specificity Matrix

A quantitative specificity matrix scoring every panel protease against every substrate motif. For each (protease, peptide) pair the model will compute a normalised cleavage score from published k_{cat}/K_M values or literature selectivity ratios, then display the result as a heatmap. The goal is to confirm that the SGRSENIRTA motif (MMP14) and the analogous motifs chosen for MMP-8, MMP-9, NE, fecal serine proteases, and FAP do not produce significant cross-cleavage within the panel. This feeds directly into the multiplexed sensor design and the Best Model prize narrative.

Task 7: ML Substrate Peptide Prediction Tool

A machine-learning pipeline to rank candidate substrate peptides for each panel protease, using ESM2 embeddings of the cleavage-site context window and a logistic classifier trained on the MEROPS database of confirmed cleavage events. The tool will output a ranked shortlist of 8-mer motifs with predicted k_{cat}/K_M and selectivity scores, giving wet-lab a shortlist for synthesis and validation. This is the primary candidate for the Best Software Tool prize.